1. 两个重要极限：① $\lim_{x\to 0}\frac{\sin x}{x}=1$　② $\lim_{x\to\infty}(1+\frac{1}{x})^x=e$ $\left[\lim_{x\to\infty}(1+\frac{a}{x})^x=e^a\right]$

2. 等价无穷小：① $x\sim\sin x\sim\tan x$　② $1-\cos x\sim\frac{1}{2}x^2$

③ $a^x-1\sim x\ln a$　④ $e^x-1\sim\ln(1+x)\sim x$　⑤ $x\sim\arcsin x\sim\arctan x$

⑥ $(1+x)^a-1\sim ax$　⑦ $\tan x-\sin x\sim\frac{1}{2}x^3$

3. 三角函数：① $\sec x=\frac{1}{\cos x}$　$\csc x=\frac{1}{\sin x}$

② $\tan^2 x=\sec^2 x-1$，$\cot^2 x=\csc^2 x-1$

4. 导数：① $(x^a)'=ax^{a-1}$　② $(\sin x)'=\cos x$，$(\cos x)'=-\sin x$

③ $(\tan x)'=\sec^2 x$，$(\cot x)'=-\csc^2 x$

④ $(\sec x)'=\sec x\tan x$，$(\csc x)'=-\csc x\cot x$

⑤ $(a^x)'=a^x\ln a$，$(e^x)'=e^x$　⑥ $(\log_a x)'=\frac{1}{x\ln a}$，$(\ln x)'=\frac{1}{x}$

⑦ $(\arcsin x)'=\frac{1}{\sqrt{1-x^2}}$，$(\arccos x)'=-\frac{1}{\sqrt{1-x^2}}$

$(\arctan x)'=\frac{1}{1+x^2}$，$(\operatorname{arccot} x)'=-\frac{1}{1+x^2}$

参数方程求导：① $\frac{dy}{dx}=\frac{\dot y}{\dot x}$　② $\frac{d^2y}{dx^2}=\frac{\dot x\ddot y-\ddot x\dot y}{\dot x^3}$

5. Taylor 展开：$f(x)=f(x_0)+f'(x_0)(x-x_0)+\frac{1}{2!}f''(x_0)(x-x_0)^2+\cdots+\frac{1}{n!}f^{(n)}(x_0)(x-x_0)^n$

$+\begin{cases}o((x-x_0)^n) & \longrightarrow \text{Peano 余项}\\ \frac{1}{(n+1)!}f^{(n+1)}(\xi)(x-x_0)^{n+1} & \longrightarrow \text{Lagrange 余项}\end{cases}$

6. Maclaurin 公式：（级数为 $+\cdots$）

① $e^x=1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}+\frac{x^{n+1}}{(n+1)!}e^{\theta x}$

② $\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots+(-1)^{n-1}\frac{x^n}{n}+(-1)^n\frac{x^{n+1}}{(n+1)(1+\theta x)^{n+1}}$

③ $\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots+(-1)^{n-1}\frac{x^{2n-1}}{(2n-1)!}+(-1)^n\frac{\cos\theta x}{(2n+1)!}x^{2n+1}$

④ $\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots+(-1)^n\frac{x^{2n}}{(2n)!}+(-1)^{n+1}\frac{\cos\theta x}{(2n+2)!}x^{2n+2}$

⑤ $(1+x)^a=1+ax+\frac{a(a-1)}{2!}x^2+\cdots+\frac{a(a-1)\cdots(a-n+1)}{n!}x^n+\cdots$

⑥ $\frac{1}{1-x} = 1+x+x^2+\cdots+x^n+\cdots$

⑦ $\frac{1}{1+x} = 1-x+x^2-\cdots+(-1)^n x^n+\cdots$

⑧ $\arctan x = x-\frac{x^3}{3}+\frac{x^5}{5}+\cdots+(-1)^n\frac{x^{2n+1}}{2n+1}+\cdots$

7. 积分：

① $\int k\,dx = kx+C$　② $\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1}+C$　③ $\int\frac{dx}{x} = \ln|x|+C$

④ $\int a^x dx = \frac{a^x}{\ln a}+C$　⑤ $\int e^x dx = e^x+C$　⑥ $\int\cos x\,dx = \sin x+C$

⑦ $\int\sin x\,dx = -\cos x+C$　⑧ $\int\tan x\,dx = -\ln|\cos x|+C$　⑨ $\int\cot x\,dx = \ln|\sin x|+C$

⑩ $\int\sec x\,dx = \ln|\sec x+\tan x|+C$　⑪ $\int\csc x\,dx = -\ln|\csc x+\cot x|+C$

⑫ $\int\sec^2 x\,dx = \tan x+C$　⑬ $\int\csc^2 x\,dx = -\cot x+C$

⑭ $\int\sec x\tan x\,dx = \sec x+C$　⑮ $\int\csc x\cot x\,dx = -\csc x+C$

⑯ $\int\frac{dx}{\sqrt{1-x^2}} = \arcsin x+C$　⑰ $\int\frac{dx}{1+x^2} = \arctan x+C$

⑱ $\int\frac{dx}{\sqrt{1+x^2}} = \ln(x+\sqrt{1+x^2})+C$　⑲ $\int\frac{dx}{1-x^2} = \frac{1}{2}\ln\frac{1+x}{1-x}+C$

⑳ $\int\sin^2 x\,dx = \frac{1}{2}(x-\sin x\cos x)+C$　㉑ $\int\cos^2 x\,dx = \frac{1}{2}(x+\sin x\cos x)+C$

㉒ $\int\ln x\,dx = x(\ln x-1)+C$　㉓ $\int xe^x dx = e^x(x-1)+C$

㉔ $\int\arcsin x\,dx = x\arcsin x+\sqrt{1-x^2}+C$　㉕ $\int\arctan x\,dx = x\arctan x-\frac{1}{2}\ln(1+x^2)+C$

8. 重要三角函数积分公式：

$$\int_0^{\frac{\pi}{2}}\sin^n x\,dx = \int_0^{\frac{\pi}{2}}\cos^n x\,dx = \begin{cases}\frac{n-1}{n}\cdots\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} & (n\text{为偶})\\ \frac{n-1}{n}\cdots\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3} & (n\text{为奇})\end{cases}$$

9. 一阶线性微分方程：

(1) $y'+P(x)y=0$　$y = Ce^{-\int P(x)dx}$

(2) $y'+P(x)y=Q(x)$　$y = e^{-\int P(x)dx}\left(\int Q(x)e^{\int P(x)dx}dx+C\right)$

(3) $\frac{dy}{dx}=f(\frac{y}{x})$ 令 $u=\frac{y}{x}$，则 $u+x\frac{du}{dx}=f(u)$

(4) Bernoulli 方程：$\frac{dy}{dx}+P(x)y=Q(x)y^a$ 令 $u=y^{1-a}$

$\Rightarrow \frac{du}{dx}+(1-a)P(x)u=(1-a)Q(x)$

10. 二阶常系数齐次方程：$\ddot{x}+a_1\dot{x}+a_2x=0 \Rightarrow \lambda^2+a_1\lambda+a_2=0$

(1) 两不同实根 λ_1,λ_2：$x=C_1e^{\lambda_1 t}+C_2e^{\lambda_2 t}$

(2) 两相等实根 $\lambda_1=\lambda_2$：$x=e^{\lambda_1 t}(C_1t+C_2)$

(3) 一对共轭复根 $\lambda_{1,2}=\alpha\pm i\beta$：$x=e^{\alpha t}(C_1\cos\beta t+C_2\sin\beta t)$

11. 二阶常系数非齐次方程：$\ddot{x}+a_1\dot{x}+a_2x=\varphi(t)e^{\mu t}$

令特解 $x^*(t)=t^ke^{\mu t}\phi(t)$

$\phi(t)=B_mt^m+B_{m-1}t^{m-1}+\cdots+B_1t+B_0$（与 $\varphi(t)$ 同次）

当 μ 不是特征根，$k=0$；当 μ 是单根，$k=1$；当 μ 是重根，$k=2$

形如 $\ddot{x}+a_1\dot{x}+a_2x=e^{\mu t}\varphi(t)\cos\nu t(\sin\nu t)$：

令特解 $x^*=t^ke^{\mu t}(Z_1(t)\cos\nu t+Z_2(t)\sin\nu t)$

当 $\mu\pm i\nu$ 不是特征根，$k=0$；是特征根，$k=1$

12. 判断可微：$\lim\limits_{\substack{x\to0\\y\to0}}\frac{f(x,y)-f(x_0,y_0)-f_x(x_0,y_0)(x-x_0)-f_y(x_0,y_0)(y-y_0)}{\sqrt{x^2+y^2}}=0$

13. 方向导数：$\frac{\partial f}{\partial \vec{l}}=\langle\nabla f,\vec{e_l}\rangle$

14. 曲线切线：$\frac{x-x_0}{\dot{x}}=\frac{y-y_0}{\dot{y}}=\frac{z-z_0}{\dot{z}}$

法平面：$\dot{x}(x-x_0)+\dot{y}(y-y_0)+\dot{z}(z-z_0)=0$

15. 曲面切平面：$F_x(P_0)(x-x_0)+F_y(P_0)(y-y_0)+F_z(P_0)(z-z_0)=0$

法线：$\frac{x-x_0}{F_x(P_0)}=\frac{y-y_0}{F_y(P_0)}=\frac{z-z_0}{F_z(P_0)}$

16. 弧长：$s=\int_\alpha^\beta \sqrt{(\dot{x}(t))^2+(\dot{y}(t))^2+(\dot{z}(t))^2}\,dt$

$=\int_a^b \sqrt{1+(y'(x))^2}\,dx$

$=\int_\alpha^\beta \sqrt{(\rho(\theta))^2+(\rho'(\theta))^2}\,d\theta$

曲率：$K(t)=\frac{\|\vec{r}(t)\times\vec{r}(t)\|}{\|\vec{r}(t)\|^3}=\frac{|\dot{x}\ddot{y}-\dot{y}\ddot{x}|}{(\dot{x}^2+\dot{y}^2)^{\frac{3}{2}}}=\frac{y''}{(1+(y')^2)^{\frac{3}{2}}}$

17. 形心：$\bar{x}=\frac{\iiint_{(v)} x\,dV}{\iiint_{(v)} dV}$ （y,z 同）

18. 线面积分：

① 一型线积分：$\int_{(c)} f(x,y,z)\,ds=\int_\alpha^\beta f(x(t),y(t),z(t))\sqrt{(\dot{x}(t))^2+(\dot{y}(t))^2+(\dot{z}(t))^2}\,dt$

② 一型面积分：$\iint_{(S)} f(x,y,z)\,dS=\iint_{(\sigma)} f(x,y,z(x,y))\sqrt{1+z_x^2+z_y^2}\,dx\,dy$

③ 二型线积分：$\int_{(c)} \vec{A(M)}\cdot\vec{ds}=\int_{(c)} P\,dx+Q\,dy+R\,dz$ $\left(\int_{(c)} P\,dx=\int_\alpha^\beta P(x(t),y(t),z(t))\dot{x}(t)\,dt\right)$

④ 二型面积分：$\iint_{(S)} \vec{A(M)}\cdot\vec{dS}=\iint_{(S)} \vec{A(M)}\cdot\vec{e_n}\cdot dS=\iint_{(S)} P\,dy\wedge dz+Q\,dx\wedge dz+R\,dx\wedge dy$

$\left[\iint_{(S)} R\,dx\wedge dy=\pm\iint_{(D_{xy})} R(x,y,z(x,y))\,dx\,dy\right.$ 符号看法线与z轴夹角$]$

19. 积分公式：

① Green 公式：$\iint_{(\sigma)}\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)d\sigma=\oint_{(+c)} P\,dx+Q\,dy$

② Gauss 公式：$\iiint_{(v)} \nabla\cdot\vec{A}\,dV=\oiint_{(\sigma)} \vec{A}\cdot\vec{dS}$

即 $\iiint_{(v)}\left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right)dV=\oiint_{(\sigma)} P\,dy\wedge dz+Q\,dy\wedge dz+R\,dx\wedge dy$

③ Stokes 公式：$\oint_{(c)} \vec{A}\cdot\vec{ds}=\iint_{(S)} \nabla\times\vec{A}\cdot\vec{dS}$

即 $\oint P\,dx+Q\,dy+R\,dz=\iint_{(S)}\begin{vmatrix} dy\wedge dz & dx\wedge dz & dx\wedge dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$

20. 四个等价命题：

① $\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}$. $\frac{\partial Q}{\partial z}=\frac{\partial R}{\partial y}$, $\frac{\partial P}{\partial z}=\frac{\partial R}{\partial x}$

② $\oint_{(C)} P\,dx + Q\,dy + R\,dz = 0$

③ 积分与路径无关

④ $P\,dx + Q\,dy + R\,dz$ 为某函数的全微分

21. 旋度散度：

① $\operatorname{div}\vec{A} = \nabla\cdot\vec{A} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$，$\operatorname{rot}\vec{A} = \nabla\times\vec{A} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$

② $\operatorname{div}(\operatorname{rot}\vec{A}) = \nabla\cdot(\nabla\times\vec{A}) = 0$ 旋度的散度为0

$\operatorname{rot}(\operatorname{grad}u) = \nabla\times(\nabla u) = \vec{0}$ 梯度的旋度为$\vec{0}$

22. Fourier 级数展开：

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx)$

$a_0 = \frac{1}{l}\int_{-l}^{l} f(x)\,dx$

$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi}{l}x\,dx$ 奇延拓 $a_n = 0$

$b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\frac{n\pi}{l}x\,dx$ 偶延拓 $b_n = 0$